BAIS:6070 Data Science | Applications Exam

Learning Goal / Big Idea / Big Question

Use supervised learning methods to make predictions.

Course Learning Objective(s)

1. Construct and evaluate various models for predicting values in a regression task.
2. Construct and evaluate various models for predicting events in a classification task.

Deliverables

Present your findings in the following formats:

- For each question, upload a single file (.docx, .pdf, .xlsx) as your response to each ICON Quiz question

Assessment and Feedback Plan

Your grade depends on the accuracy and completeness of your analysis as well as the clarity of your interpretations. Feedback will be provided via SpeedGrader in the corresponding ICON Assignment dropbox.

Grading Breakdown:

Question 1: 50 points
Question 2: 60 points
Question 3: 30 points
Question 4: 30 points
Question 5: 30 points

Description

It is summer of 2019 and your team in the National Football League (NFL) marketing department has been assigned to construct a set of models for forecasting the national television viewership for each game of the upcoming Monday Night Football season in the autumn of 2019.

The file *MNF.csv* contains data on Monday Night Football games from 1991 to 2018. The variables tracked are:

Game	Variable indexing order of games
Network	Television network broadcasting game
AwayTeam	NFL team that is visitor for game
HomeTeam	NFL team hosting game
Date	Calendar date of game

Viewership	Standardized television viewership measure with larger positive values representing more viewership (target variable for regression task)
PoorGame	Whether viewership of game achieved low viewership (target variable for classification task)
Week	Index of the week in NFL season that game occurred
ElectionYear	Whether game occurred in a year with a U.S. Presidential election
MaxScoreAvg	Normalized scoring average from previous season of team with higher scoring average of two teams in game
MinScoreAvg	Normalized scoring average from previous season of team with lower scoring average of two teams in game
BestFranchiseRank	Forbes' franchise valuation rank of team with highest value of two teams in game (smaller value = higher rank)
WorstFranchiseRank	Forbes' franchise valuation rank of team with lowest value of two teams in game (larger value = lower rank)
EliteQBs	Number of quarterbacks in game considered to be elite
MaxPrevWins	Number of wins from previous season of team with largest number of wins of two teams in game
MinPrevWins	Number of wins from previous season of team with smallest number of wins of two teams in game
PlayoffRematch	Whether game is a rematch of playoff game from previous season
Divisional	Whether game is between divisional opponents
SuperBowlChamp	Whether game involves the previous season's Super Bowl champion
WorstOddsSB	Preseason odds (payout for $100 bet) of winning the Super Bowl for the team least likely of two teams in game
BestOddsSB	Preseason odds (payout for $100 bet) of winning the Super Bowl for the team most likely of two teams in game

As you construct your prediction models, be mindful to explore the data using statistical summaries and data visualization to gain insight. In addition to the variables supplied, this exploratory data analysis may uncover other features to engineer that may provide

predictive value. Be sure to investigate any missing data and the impact of how handling it affects your predictive analysis.

1. Construct a regression model to predict the value of a game's viewership (target variable: Viewership). Use the entire data set (100%) in a 10-fold cross-validation experiment. That is, do not set any data aside as a test set.
    a) Use 10-fold cross-validation to evaluate different predictive models and select a final model, reporting performance measures from the cross-validation experiment. Given that we typically do not know what the best possible prediction performance for a task is, we must benchmark our efforts by examining several competing approaches, including:
        i. Lasso linear regression
        ii. k-nearest neighbors regression
        iii. Random forest regression
        iv. Gradient boosting regression
        v. Neural network regression
    b) Construct a final model on the entire data set and generate the corresponding predictions of Viewership for the 17 games in the 2019 season provided in *2019games.csv*. Note that any manipulation and pre-processing of the features used in the data to construct the prediction model must also be performed on the data in *2019games.csv* prior to executing your predictions on these 19 games.

2. Construct a classification model to predict whether a game will be exceptionally poor relative to that season's viewership using PoorGame as the target variable. Use the entire data set (100%) in a 10-fold cross-validation experiment. That is, do not set any data aside as a test set.
    a) Use 10-fold cross-validation to evaluate different predictive models and select a final model, reporting performance measures from the cross-validation experiment. Given that we typically do not know what the best possible prediction performance for a task is, we must benchmark our efforts by examining several competing approaches, including:
        i. Lasso logistic regression
        ii. k-nearest neighbors classifier
        iii. Random forest classifier
        iv. Gradient boosting classifier
        v. Neural network classifier
        vi. Support vector machine classifier
    b) Construct the final model on the entire data set and generate the predicted PoorGame probability for each of the 17 games in the 2019 season provided in *2019games.csv*. Note that any manipulation and pre-processing of the features used in the data to construct the prediction model must also be

performed on the data in *2019games.csv* prior to executing your predictions on these 17 games.

3. Submit a file called Predictions_YourName.xlsx (replacing <YourName> with your actual name) that appends two columns to the game data in the 2019games.csv file: the first appended column should contain your predicted values of from 1(b with a column title of Viewership <Your Name> and the second appended column should contain your predicted PoorGame probabilities from 2(b) with a column title of PoorGame <Your Name>.

4. Compare the regression viewership predictions in 1(b) to the classification PoorGame probabilities in 2(b). Are these predictions in agreement? If not, for which 2019 games do they seem to disagree?

5. The NFL and its broadcast partners are interested in identifying games for the 2019 season that may be "poor games" because companies who have purchased advertising time for these games often are placated with complimentary advertising time within a few days of the game with low viewership. Knowing the potential for "poor games" allows the broadcast partners to reserve an appropriate amount of advertising time to compensate for these games. Reserving advertising time within a time window of a few days after a game for poor-game compensation costs a broadcast partner $400,000 per game. However, if no advertising time is reserved and a poor game occurs, broadcast partners estimate that the financial impact is $1,200,000 per game. Which of the 2019 games would you classify as a "poor game?"